Table 1. Data sources

Dataset / endpoint	Cases	Controls	Exposure or clinical role
eQTLGen blood expression	—	—	Exposure: up to 31,684; predominantly European
BBJ GD (GCST90018627)	2,809	172,656	Discovery; East Asian
FinnGen GD (E4_GRAVES_STRICT)	3,962	496,386	Post hoc GD comparison; Finnish
UKB hyperthyroidism (GCST90038636)	3,731	480,867	Self-report; ancestry not established as exclusively European
FinnGen GO (GRAVES_OPHT)	858	499,490	Original ophthalmopathy comparison; Finnish
FinnGen GO strict (E4_GRAVES_OPHT_STRICT)	753	499,595	Post hoc definition sensitivity; Finnish

GD, Graves disease; GO, Graves ophthalmopathy; HT, hyperthyroidism/thyrotoxicosis. The three FinnGen R12 endpoints overlap and use population non-cases for their respective definitions. Strict is the source endpoint label, not proof of perfect specificity. The descriptive orbital series (four TED patients, one control) is separate from the genetic datasets.

Table 2. TSHR, IGF1R and CTLA4 across five disease datasets

Gene	Dataset	IVs	OR (95% CI)	P value	PP.H3	PP.H4
TSHR	BBJ GD	1	0.12 (0.07–0.21)	**1.09×10^{-14}**	0.049	**0.951**
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	**3.19×10^{-32}**	0.736	0.264
TSHR	UKB HT	1	0.09 (0.06–0.14)	**8.77×10^{-28}**	0.774	0.226
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	**2.82×10^{-7}**	0.014	**0.986**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	**9.02×10^{-5}**	0.082	**0.891**
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	**0.0212**	0.236	0.073
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.259	0.010
IGF1R	UKB HT	4	1.35 (1.07–1.70)	**0.0117**	0.196	0.404
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.346	0.032
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.391	0.038
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	**5.45×10^{-15}**	0.799	0.201
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	**2.15×10^{-6}**	0.746	0.254
CTLA4	UKB HT	2	0.21 (0.08–0.57)	**0.0022**	0.047	**0.953**
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	**0.0118**	0.022	**0.978**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	**0.00888**	0.021	**0.979**

IVs, instrumental variables (retained genetic variants). Wald ratio for one IV; random-effects IVW for multiple IVs. ORs are per approximately one SD higher reconstructed blood expression. Bold P values denote nominal $P < 0.05$ only. Discovery uses 0.05/2,544. PP.H3 supports distinct variants; PP.H4 supports one shared variant. Bold PP.H4 denotes ≥0.80 under $p12 = 10^{-5}$. These are probabilities, not P values. CTLA4 multi-IV results are concentrated in rs13030124; TSHR is single-IV in all datasets.

Table 3. Screen and supporting analyses — A

BBJ discovery gene	BBJ OR (95% CI)	BBJ P	FinnGen GD OR (95% CI)	FinnGen GD P	Focused support	GD PP.H4
HLA-A	1.98 (1.73–2.27)	**2.60×10^{-23}**	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	2.39 (1.95–2.93)	**8.30×10^{-17}**	1.90 (1.64–2.20)	**1.41×10^{-17}**	Yes	<0.001
CTLA4	0.18 (0.11–0.27)	**5.45×10^{-15}**	0.17 (0.08–0.36)	**2.15×10^{-6}**	Yes	0.254
TSHR	0.12 (0.07–0.21)	**1.09×10^{-14}**	0.08 (0.05–0.12)	**3.19×10^{-32}**	Yes	0.264
C4A	0.45 (0.35–0.57)	**2.23×10^{-10}**	0.41 (0.26–0.63)	**4.97×10^{-5}**	Yes	<0.001
HSD3B7	0.29 (0.18–0.47)	**2.04×10^{-7}**	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	1.49 (1.28–1.74)	**3.14×10^{-7}**	3.28 (2.09–5.17)	**2.68×10^{-7}**	Yes	<0.001
VKORC1	0.13 (0.06–0.29)	**6.35×10^{-7}**	NA	NA	NA	0.011
TNFSF14	0.63 (0.52–0.76)	**1.48×10^{-6}**	0.77 (0.65–0.92)	**0.00361**	Yes	0.303
PRSS36	6.00 (2.87–12.57)	**2.02×10^{-6}**	1.40 (1.00–1.96)	**0.049**	No	0.063
MAPKAPK5	0.977 (0.967–0.987)	**5.28×10^{-6}**	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	1.25 (1.13–1.38)	**6.86×10^{-6}**	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	2.10 (1.51–2.91)	**9.41×10^{-6}**	1.62 (1.18–2.24)	**0.0033**	Yes	0.313

Panel A: focused support requires concordant direction and $P < 0.05/13$; 7 of 13 qualify. NA is unavailable, not a negative result. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.

Table 3. Screen and supporting analyses — B

Additional gene	GD IVs	GD OR (95% CI)	GD P	BBJ PP.H4	GD PP.H4	GO strict PP.H4
APOM	1	0.05 (0.04–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.227
NEU1	1	0.04 (0.03–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.061
IGLV2-18	1	0.56 (0.48–0.66)	**7.52×10^{-12}**	0.029	0.020	0.100
IGLV3-12	1	1.20 (1.14–1.26)	**7.52×10^{-12}**	0.019	0.022	0.101
TNF	1	0.60 (0.51–0.71)	**4.65×10^{-10}**	<0.001	<0.001	<0.001
HIST1H4E	1	2.54 (1.88–3.43)	**1.51×10^{-9}**	<0.001	<0.001	0.007
LTA	1	0.46 (0.35–0.61)	**3.75×10^{-8}**	<0.001	<0.001	<0.001
LTB	1	0.18 (0.10–0.36)	**4.89×10^{-7}**	<0.001	<0.001	<0.001
C4B	2	1.63 (1.31–2.03)	**1.10×10^{-5}**	<0.001	<0.001	0.311

nine additional reference-frequency FinnGen GD discoveries at $P < 0.05/2{,}544$; post hoc and not independent of the full screen. APOM/NEU1 share rs3101018; IGLV2-18/IGLV3-12 share rs5759410. The lead IGLV expression variant is absent in BBJ, weakening interpretation of its low PP.H4.

Table 3. Screen and supporting analyses — C

Gene / GD analysis	Reference OR (95% CI)	Reference P	Cohort-frequency OR (95% CI)	Cohort P	Interpretation
C4B	1.63 (1.31–2.03)	**1.10×10^{-5}**	1.75 (1.31–2.33)	**1.48×10^{-4}**	Heterogeneous; loses screen significance
HLA-G	0.86 (0.72–1.03)	0.107	0.60 (0.47–0.75)	**1.72×10^{-5}**	Weight-sensitive; single-variant dependence

HLA-G was added because of frequency sensitivity, not primary discovery. Neither C4B nor HLA-G gains strong GD colocalization. All 24 genes and five datasets are available in the workbook. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.

Figure 1. Study design and BBJ discovery

A Blood expression instruments
4,462 druggable genes considered
2,544 genes; 6,135 SNP–gene records

→

BBJ GD discovery
2,234 estimable genes; 13 discoveries
Threshold P < 0.05/2,544

→

Disease-definition comparisons
FinnGen GD, UKB hyperthyroidism
Two overlapping GO definitions

Regional follow-up: TSHR and IGF1R at the centre; CTLA4 comparator; 24 selected genes in total.

B BBJ gene-level MR associations

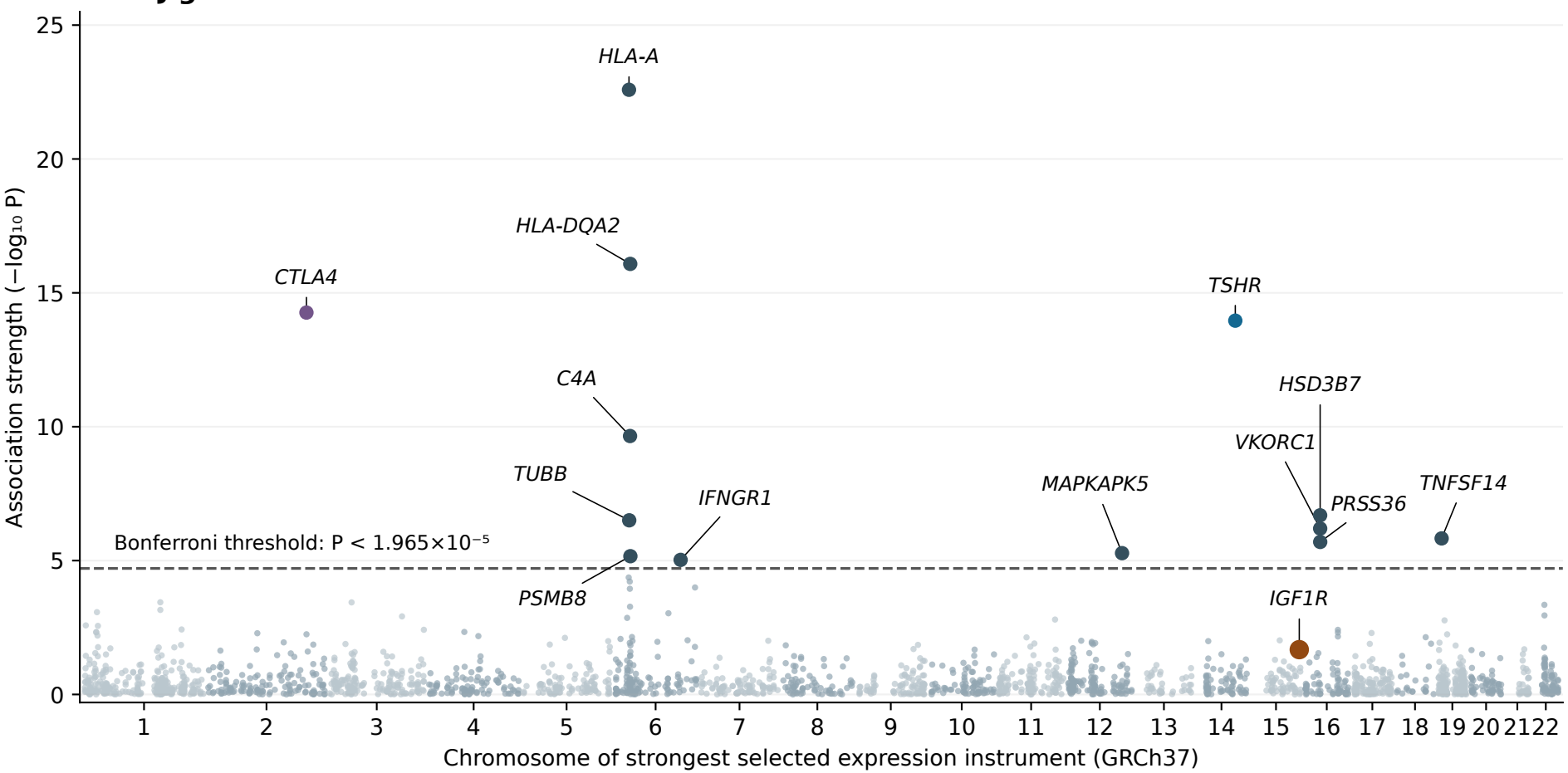


Figure 2. TSHR and IGF1R across disease definitions, with CTLA4 as comparator

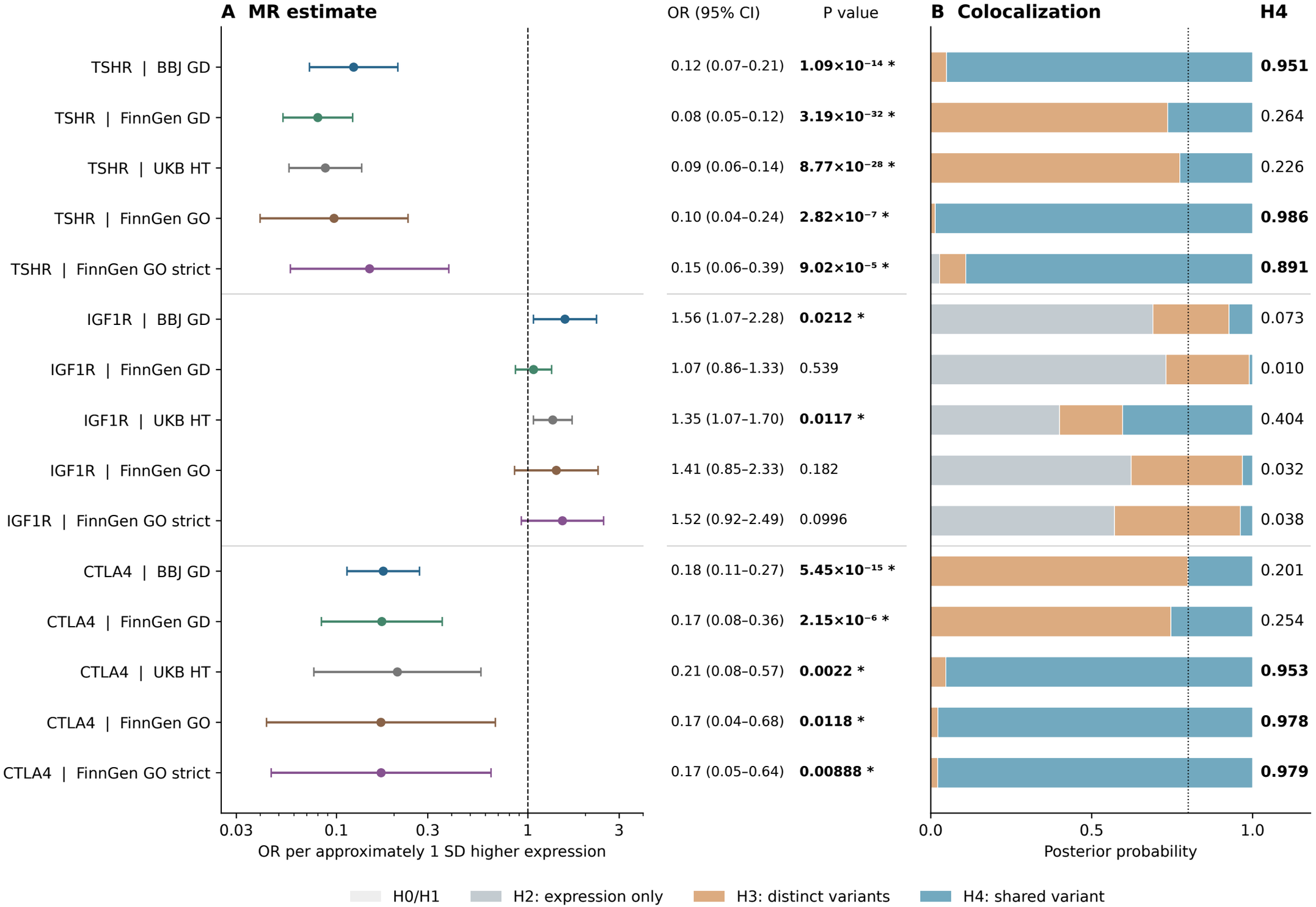


Figure 3. Regional genetic evidence and TSHR prior sensitivity

A Shared-variant probability (PP.H4)

	BBJ GD	FinnGen GD	UKB HT	FinnGen GO	FinnGen GO strict
TSHR	0.951	0.264	0.226	0.986	0.891
IGF1R	0.073	0.010	0.404	0.032	0.038
CTLA4	0.201	0.254	0.953	0.978	0.979
HLA-A	<0.001	<0.001	<0.001	<0.001	<0.001
HLA-DQA2	<0.001	<0.001	<0.001	<0.001	<0.001
C4A	<0.001	<0.001	<0.001	0.030	0.326
HSD3B7	0.636	0.011	0.095	0.026	0.046
TUBB	<0.001	<0.001	<0.001	<0.001	<0.001
VKORC1	0.388	0.011	0.030	0.030	0.048
TNFSF14	0.994	0.303	0.237	0.017	0.018
PRSS36	0.168	0.063	0.115	0.037	0.033
MAPKAPK5	<0.001	0.041	0.005	0.026	0.029
PSMB8	<0.001	<0.001	<0.001	<0.001	<0.001
IFNGR1	0.989	0.313	0.052	0.020	0.021
APOM	<0.001	<0.001	<0.001	0.006	0.227
NEU1	<0.001	<0.001	<0.001	0.003	0.061
IGLV2-18	0.029	0.020	0.024	0.082	0.100
IGLV3-12	0.019	0.022	0.024	0.083	0.101
TNF	<0.001	<0.001	<0.001	<0.001	<0.001
HIST1H4E	<0.001	<0.001	<0.001	0.004	0.007
LTA	<0.001	<0.001	<0.001	<0.001	<0.001
LTB	<0.001	<0.001	<0.001	<0.001	<0.001
C4B	<0.001	<0.001	<0.001	0.030	0.311
HLA-G	<0.001	<0.001	<0.001	<0.001	0.001

B TSHR prior sensitivity

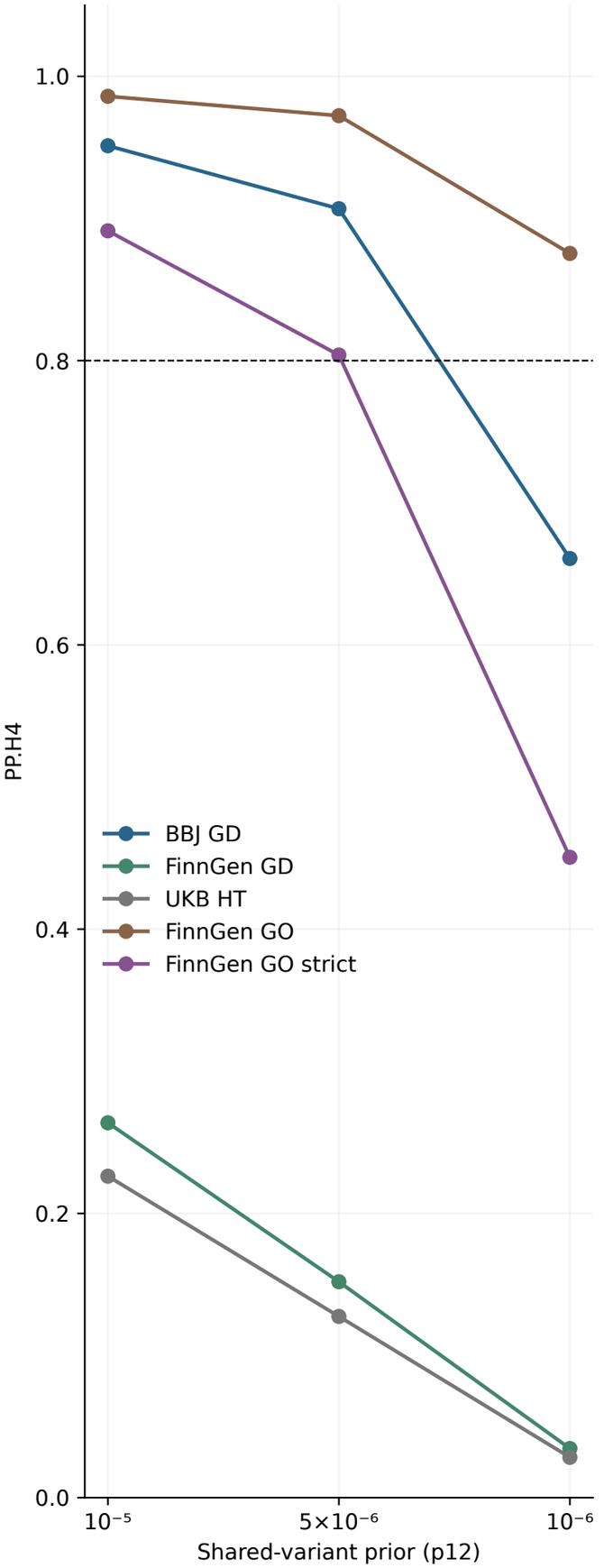


Table S1. TSHR regional and prior sensitivity

Dataset	Full cis SNPs	Primary PP.H4	Lower-prior PP.H4	PP.H4 range across windows
BBJ GD	4336	**0.951**	0.661	0.951–0.951
FinnGen GD	6579	0.264	0.035	0.264–0.264
UKB HT	5667	0.226	0.028	0.226–0.226
FinnGen GO	6579	**0.986**	**0.875**	0.986–0.987
FinnGen GO strict	6579	**0.891**	0.451	0.891–0.905

Table S1. TSHR regional and prior sensitivity. Lower prior: p12 = 10^{-6}; p1 = p2 = 10^{-4} throughout. Window range uses the full source cis-region, ±500 kb, ±250 kb and five-outcome intersection at the primary prior. It is not a confidence interval. Full regional disease lead variants were retained. All 240 settings are in TSHR_windows.

Table S2. Frequency-sensitive exploratory candidates

Gene / dataset	Frequency	IVs	OR (95% CI)	P value	Q-test P
C4B / FinnGen GD	Reference	2	1.63 (1.31–2.03)	**1.10×10^{-5}**	7.79×10^{-7}
C4B / FinnGen GD	eQTLGen	2	1.75 (1.31–2.33)	**1.48×10^{-4}**	1.36×10^{-8}
C4B / FinnGen GO strict	Reference	2	1.68 (1.44–1.95)	**2.74×10^{-11}**	0.122
C4B / FinnGen GO strict	eQTLGen	2	1.80 (1.46–2.23)	**6.17×10^{-8}**	0.0587
HLA-G / FinnGen GD	Reference	16	0.86 (0.72–1.03)	0.107	9.98×10^{-24}
HLA-G / FinnGen GD	eQTLGen	16	0.60 (0.47–0.75)	**1.72×10^{-5}**	2.26×10^{-10}
HLA-G / FinnGen GO strict	Reference	16	0.91 (0.76–1.08)	0.267	0.0231
HLA-G / FinnGen GO strict	eQTLGen	16	0.70 (0.53–0.92)	**0.0122**	0.131

Table S2. C4B and HLA-G frequency sensitivity. Variant sets and outcome effects were held fixed. Frequency changes rescaled expression estimates and IVW weights. Q tests heterogeneity, not directional pleiotropy; its P value is not a gene-association P value. All 72 omission estimates and instrument weights are provided in the workbook.

Table S3. Tissue-specific expression and FinnGen GD

Gene / tissue	N	IVs	OR (95% CI)	P value	Adjusted P
TSHR / Thyroid	574	0	No qualifying IV	—	—
IGF1R / Thyroid	574	1	0.76 (0.56–1.02)	0.0692	0.692
TSHR / Cultured fibroblasts	483	0	Not tested	—	—
IGF1R / Cultured fibroblasts	483	1	1.11 (0.89–1.38)	0.3591	1.000
TSHR / Subcutaneous adipose	581	1	0.36 (0.31–0.43)	1.28e-34	1.28e-33
IGF1R / Subcutaneous adipose	581	0	No qualifying IV	—	—
TSHR / Visceral adipose	469	1	0.50 (0.43–0.58)	4.09e-19	4.09e-18
IGF1R / Visceral adipose	469	0	No qualifying IV	—	—
TSHR / Whole blood	670	0	No qualifying IV	—	—
IGF1R / Whole blood	670	0	No qualifying IV	—	—

Table S3. Fixed-panel tissue extension with FinnGen GD as the primary disease dataset. IVs are retained instrumental variants. ORs are per unit of normalized, inverse-normal-transformed expression. Adjusted P = min(1,10P); the ten-comparison denominator includes unestimable cells. Not tested means the gene was absent from the tissue eGenes file; no qualifying IV means the minimum eQTL P did not meet 5×10^{-8}. All four estimable combinations used one SNP (F 32.1–61.1). The 40 secondary comparisons use a separate denominator of 40 and are retained, with all source statistics, in Tissue_extension. Cultured fibroblasts and adipose tissues are not orbital tissue. Original detection benchmarks remain in Detection_original.

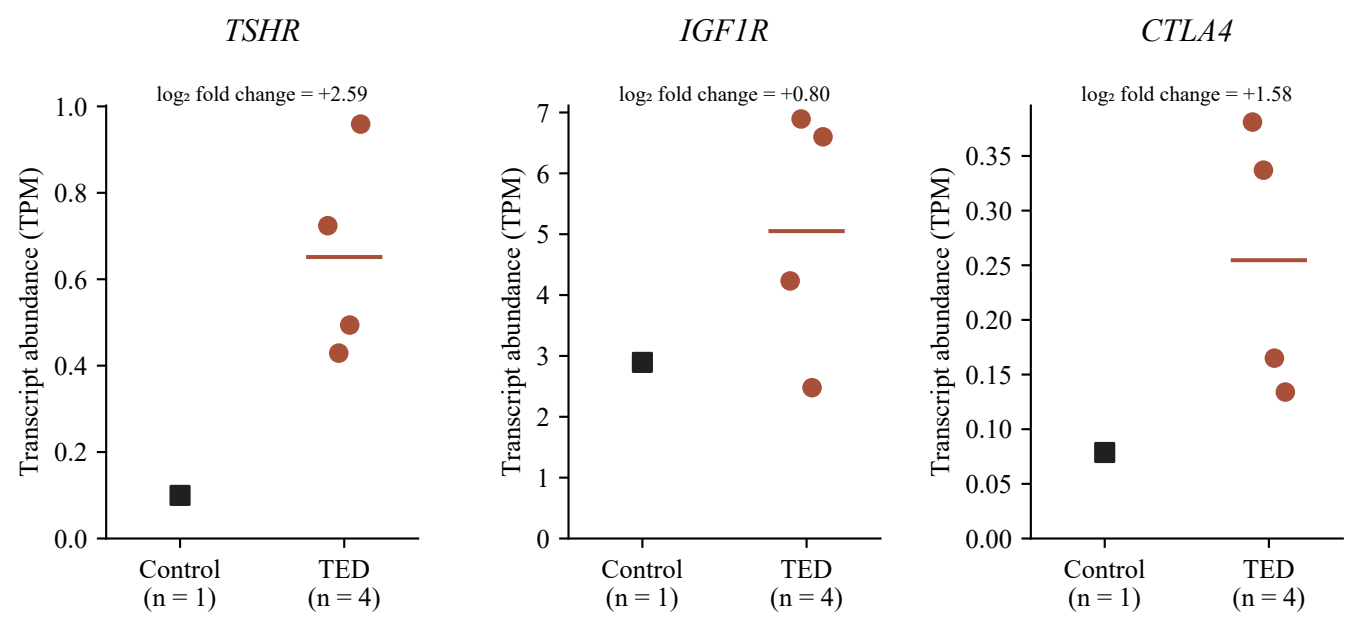
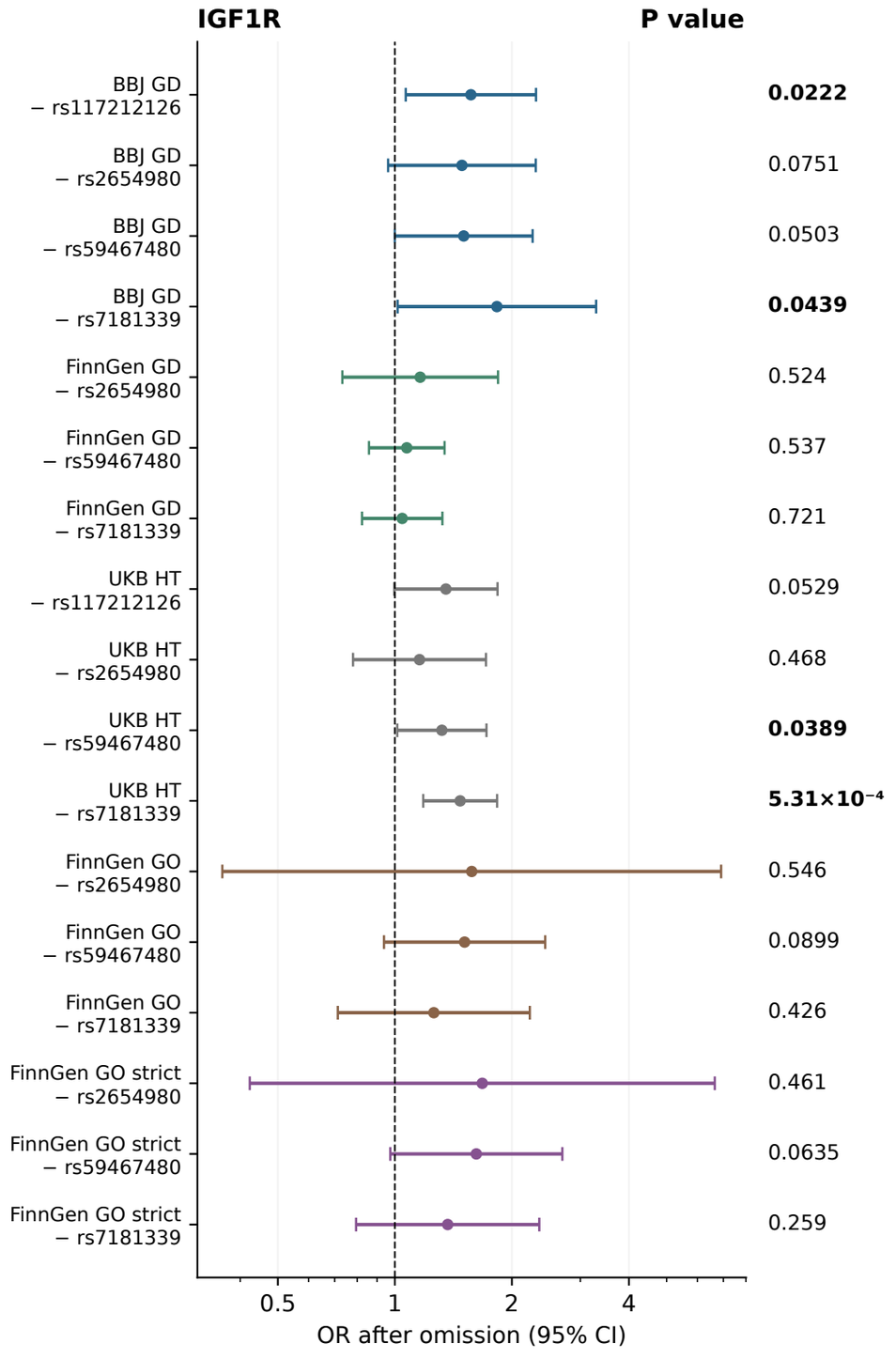
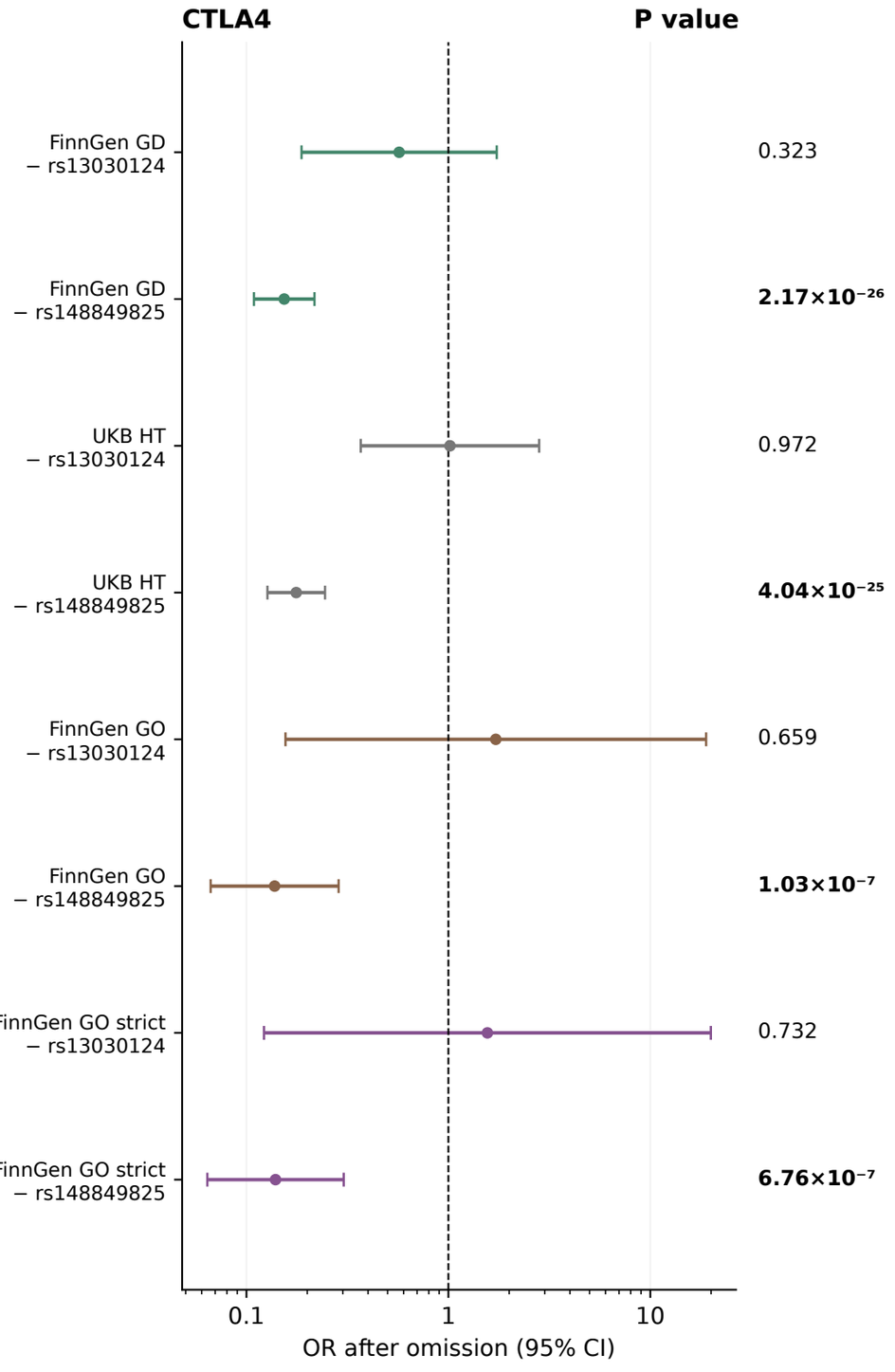

TSHR
log₂ fold change = +2.59
Transcript abundance (TPM)
1.0
0.8
0.6
0.4
0.2
0.0
Control
(n = 1)
TED
(n = 4)
IGF1R
log₂ fold change = +0.80
Transcript abundance (TPM)
7
6
5
4
3
2
1
0
Control
(n = 1)
TED
(n = 4)
CTLA4
log₂ fold change = +1.58
Transcript abundance (TPM)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Control
(n = 1)
TED
(n = 4)

Figure S2. Sensitivity of IGF1R and CTLA4 estimates to individual instruments





Each row omits the named SNP. Original reference frequencies; 25 correlated omission estimates.
TSHR and BBJ CTLA4 are single-instrument analyses and cannot undergo omission.